1. Clinical Question & Core Hypothesis

Clinical Question

TLS-rich immune organization is often associated with better anti-tumor immunity, yet many NMIBC tumors that display lymphoid aggregates still fail intravesical BCG therapy—why does organized local immunity not always translate into durable therapy response? 2) Tier 1 paths used: I built on the PDCD1 (PD-1/PD-L1) path because BCG can induce PD-L1 and thereby engage T cell inhibitory signaling, and on the VEGFA/BMAL1 angiogenesis path because VEGF-driven vascular remodeling is a well-characterized mechanism that creates immune-excluding and immunosuppressive microenvironments; together these provide a plausible mechanism linking TLS function to therapy failure. 3) Tier 2 support: EP300, HCFC2 and TNPO3 (circRNA) supply epigenetic and post-transcriptional regulatory mechanisms that could modulate PD-L1/PDCD1 and VEGFA programs; AURKA and TP53/KRAS context suggest cell-cycle and oncogenic pressures that may further blunt TLS efficacy. 4) Why therapy_response endpoint: BCG response is the most clinically actionable, mechanistically linked outcome in NMIBC (the PD-L1 induction path centers on BCG), and is more sensitive than OS for testing microenvironment-driven mechanisms in this setting. 5) Deliberate exclusions: I excluded using KRAS as a core anchor to keep the hypothesis mechanistically focused on immune–vascular crosstalk; other Tier 2/Tier 3 genes and broad signatures (79-gene, stem-like, micropapillary) were relegated to secondary_modifiers to avoid overcomplicating the core causal question?

Working Hypothesis

Primary hypothesis: In NMIBC, TLS-like lymphoid aggregates are not universally associated with favorable BCG response; in TLS-positive tumors with high VEGFA expression (VEGFAhigh), VEGF-driven vascular remodeling creates an immunosuppressive microenvironment that promotes PD-1/PD-L1–mediated T cell dysfunction within the TLS, leading to reduced BCG efficacy. This results in a distinct non-responder subset defined by TLS presence, VEGFAhigh status, and elevated PDCD1/PD-L1 signaling on intratumoral immune cells. Predictions: (1) TLS+VEGFAhigh tumors will exhibit increased PDCD1 expression on CD8+ T cells and other exhausted/exhausted-like states within TLS, reduced effector function (lower granzyme B/IFN-γ), and diminished BCG response compared with TLS+VEGFAlow or TLS− tumors. (2) Spatial analyses will show immature or dysfunctional TLS architecture with abnormal vasculature (abnormal vessel density, pericyte detachment) and immune exclusion in TLS+VEGFAhigh tumors. (3) Epigenetic/post-transcriptional regulators (EP300, HCFC2, TNPO3) will modulate PD-L1/PDCD1 axis and VEGFA signaling, such that perturbations elevating EP300/HCFC2/TNPO3 activity correlate with higher PD-L1 and VEGFA expression and poorer BCG response. (4) Oncogenic context (AURKA, TP53, KRAS) will modulate TLS maturation and sensitivity to vascular/ checkpoint perturbations, with AURKA amplification and TP53 loss associated with more suppressive TLS microenvironments. (5) Therapeutically, combining intravesical BCG with PD-1/PD-L1 blockade and anti-angiogenic modulation will restore TLS integrity, enhance CD8+ T cell activity within TLS, and improve therapy response in TLS+VEGFAhigh tumors. Biomarker signature for prediction: TLS maturity status (presence of germinal centers, follicular dendritic cell networks), VEGFA expression level, and PDCD1/PD-L1 axis activity on intratumoral immune cells.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PDCD1, VEGFA
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PDCD1: PDCD1 (PD-1/PD-L1) path because BCG can induce PD-L1 and thereby engage T cell inhibitory signaling, and on the VEGFA/BMAL1 angiogenesis path because VEGF-driven vascular remodeling is a well-characterized mechanism that creates immune-excluding and immunosuppressive microenvironments; together these provide a plausible mechanism linking TLS function to therapy failure.

VEGFA: VEGFA/BMAL1 angiogenesis path because VEGF-driven vascular remodeling is a well-characterized mechanism that creates immune-excluding and immunosuppressive microenvironments; together these provide a plausible mechanism linking TLS function to therapy

failure.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	BMAL1	Supporting modifier from knowledge graph
Secondary	EP300	Supporting modifier from knowledge graph
Secondary	AURKA	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	KRAS	Supporting modifier from knowledge graph
Secondary	TNPO3	Supporting modifier from knowledge graph
Secondary	HCFC2	Supporting modifier from knowledge graph
Secondary	PROS1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: can tune PD-L1/PDCD1 and VEGFA programs, while oncogenic drivers (AURKA, TP53, KRAS) shape cell-cycle and immune editing pressures that influence TLS maturation.

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- BMAL1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- EP300: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.